

Affordances in UX Design

Highlights

- The concept of affordance
- Five different types of affordances and how each kind is used in UX design
- Cognitive affordance
 - How shared conventions affect the interpretation of cognitive affordances
 - False cognitive affordances
- Physical affordance
- Sensory affordance
- Functional affordance
- Emotional affordance
- Putting affordances together in UX design
- A UX design checklist of affordances
- User-created affordances as a wake-up call to designers

27.1 INTRODUCTION

27.1.1 Acknowledgement of Source

To begin, we gratefully acknowledge a paper in *Behaviour & Information Technology* (Hartson, 2003), published by Taylor & Francis, Ltd., of London, as the primary source of the material of this chapter.

27.1.2 The Concept of Affordance

The relevant part of what the dictionary says about “to afford” is that it means to offer, yield, provide, give, or furnish something. For example, a particular window in a house may afford a fine view of the outdoors; the window helps one see that nice view. In UX design, where we focus on helping the user, an affordance is something that helps the user understand, do, sense, or feel something.

See [Section 27.9](#) for some history related to the concept of affordances.

27.1.3 Confusion Over Affordances in Early Human–Computer Interaction and UX

Norman (1990) rightly thought that this idea of affordance was useful in usability and interaction design. The problem was, however, that people were using the term very loosely in the human–computer interaction (HCI) literature. It was becoming overused, misused, confused, and abused. As a result, the concept was not working well as part of our common ground to communicate efficiently and precisely.

Preamble to Norman’s 1999 article. Norman, seeing this confusion, got upset and, in good Norman fashion, he decried this situation in his 1999 article: “I was lurking in the background of a CHI-Web discussion, when I lost all reason: I just couldn’t take it anymore. ‘I put an affordance here and an affordance there,’ a participant would say, ‘I wonder if that object affords clicking.’ ‘Affordances this, affordances that...’ ‘No!’ I screamed, and out came this note.”

In hopes of clarifying the distinctions among types of affordances, Norman came up with terms such as “real affordances” and “perceived affordances” and tried to tie them to the concepts of Gibson (1977), with whom he had spent some time in discussion.

However, it seems that his attempt to clarify only muddled the waters. People still failed to get it, and they continued to use “affordance” with little shared understanding.

Hartson’s taxonomy of affordances. So, I decided to try my hand at it and wrote a paper using terminology that I thought could help HCI and UX practitioners correctly identify and use the different kinds of affordances (Hartson, 2003). I tried to connect affordances to human-centered interaction concepts, to the things we humans do as we interact with any kind of machine—our sensory actions, our cognitive actions, and our physical actions. I also thought it was important to show how you can think of interaction design in terms of the affordances that go with these different kinds of actions. So far, the ideas in this article have held up well over time, and they are the foundation for this chapter.

27.1.4 Examples of How Cognitive Affordances Can Be Informed by Shared Cultural Conventions

Mailboxes as a symbolic cognitive affordance. One of your authors, Pardha Pyla, grew up in India, and when he would see early software computer systems that used an icon like that shown in [Fig. 27.1](#) for a mail program, he would wonder, “What is that thing?” Most homes in India do not have such a mailbox; in



Fig. 27.1

Mailboxes are common in the United States.

India, the postman knocks (twice?) on the door and delivers the mail directly to the occupant. Even when places had an external mailbox, they did not look like the image in the figure. Therefore he had no context for sharing that convention.

Only when Pardha got to the United States did he see these things, and, upon close investigation, realized they are mailboxes.

Exit signs in the United States vs. Europe. The pictures in [Fig. 27.2](#) show two different kinds of exit signs.

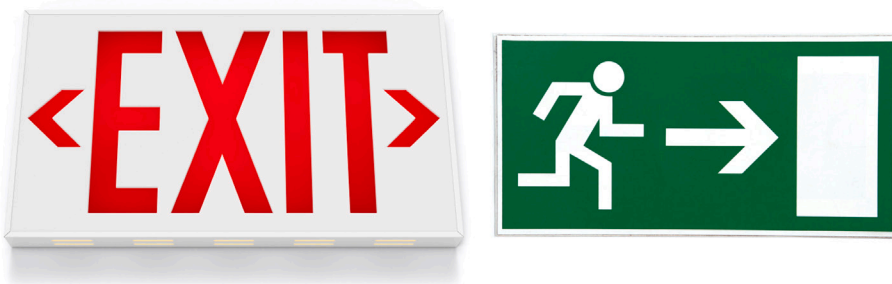


Fig. 27.2

Exit signs from two different cultures.

The one on the left has shared meaning in the United States. It is bright red, signifying danger (although maybe an exit should represent safety in the face of danger, but never mind). It has arrows indicating the direction(s) to go.

However, in Europe and some other parts of the world, they use something like the sign on the right-hand side of the figure. Its meaning seems fairly clear on the surface, but in the United States, a panicked person in a state of emergency may find it unfamiliar and possibly confusing.

Relying on the inherent characteristics of a Coke bottle. As we said, a shared cultural understanding often conveys much of the meaning of cognitive affordances. In the absence of an applicable cultural convention, we are left to rely solely on deducing meaning from the intrinsic characteristics of an object.

What do you think of the very familiar shape of a Coke bottle? What are some affordances that you can think of that might go with it? Just by looking at it, what do you think you can do with it? This includes good physical affordance to grip it, the functional affordance to hold liquid to drink, and possibly a nice aesthetic look as an emotional affordance. All of these are relative to our own experience in seeing, holding, and using these bottles for their intended purpose—a container for a refreshing drink.

In a graduate class of UX design students, we raised the question of how some of our most reliable cognitive affordances (e.g., that indicate how to operate a doorknob) may be perceived by people “from another planet,” who did not have the same shared conventions. We handed out identical empty Coke bottles to several groups and asked them to look at the bottles, to hold and handle them, and to think about what kinds of uses the inherent affordances evoke. We wanted them to get down to Gibson’s (1977) ecological level.

Students responded with the usual answers about affordances evinced by sight and touch. Visually, a Coke bottle has obvious affordances as a vessel to hold something (e.g., flowers), or it can serve as a device for rough liquid volume measurements. The heft and sturdiness sensed when held in the hand indicate affordances to serve as a paperweight, a plumb bob on a string, or even an oddly shaped rolling pin.

But what if you’ve never seen or heard of a Coke bottle before? Then how do you figure out what it is and what it is used for? That is the basis for the movie *The Gods Must Be Crazy*.

It opens with the pilot of a small plane flying over Southern Africa. He is drinking the last of a Coke, and he throws the empty bottle out the window. Down and down it floats. As this puzzling artifact from another culture lands at the feet of a Bushman in the deep Kalahari, the scene deals directly with affordances. The Bushman is seeing a Coke bottle for the first time. Naturally, he takes it back to his village to show his friends. To the villagers, it is a strange and beautiful thing, but they are puzzled about what it could be used for (its functional affordances) and how to use it (the cognitive affordances).

Its appearance triggers a curious exploration of the possibilities of this object within their own Kalahari culture. This part of the movie makes the point about ecological affordances so well. Because the Kalahari Bushmen have never seen a Coke bottle (or any drink bottle) in their lives, there is no shared convention about its meaning—namely, what it is used for. They can rely only on its inherent characteristics as clues to deduce its possible uses, without influence from cultural conventions or practice. These characteristics are pure ecological cognitive affordances, the intrinsic, raw, Gibson-style affordances of an object in its environment.

Their ecological exploration led to interesting and sometimes amusing effects. One guy got his finger stuck in the top, and the children thought that was funny. By blowing over the opening, some discovered it could be an entertainment device with which they could make odd sounds and music. After much exploration they saw the smooth, hard, and curved surface was perfect for curing snake skins. Perhaps it could even be used as a weapon to attack one another.

Every day they discovered a new use for the thing sent by the gods. One obvious use was to transport water. Its efficacy in mashing soft roots and other vegetables and grinding grain made it a most useful labor-saving device. You can learn quite a bit about affordances by watching Kalahari Bushmen explore a newfound Coke bottle.

These affordances became apparent to the Bushmen but may not be obvious to most people from an industrialized part of the world. This indicates the impact of social experiences and cultural conventions on influencing, and even prejudicing, our perceptions of an object's affordances.

27.1.5 The Importance of Affordance Issues in UX Design

Few topics in HCI or UX have been as misunderstood and misused as the notion of affordance. Yet few concepts are that central to effective UX design. In fact, the concept of affordance is one of the most fundamental notions pervading all of UX design. Because affordances are all about helping users do things, you can imagine that the majority of usability and UX problems stem from designs that do not help the user as much as they should. Likewise, the majority of UX design guidelines are also about affordances.

27.1.6 Five Different Kinds of Affordance in UX Design

Each type of affordance plays a different role in supporting users during interaction, each reflecting user processes and the kinds of actions users make in task performance.

Table 27.1
Summary of affordance types

Affordance type	Description	Example
Cognitive affordance	Design feature that helps users with their cognitive actions: thinking, deciding, learning, remembering, and knowing about things	A button label that helps users know what will happen if they click on it
Physical affordance	Design feature that helps users with their physical actions: clicking, touching, pointing, gesturing, and moving things	A button that is large enough so that users can click on it accurately
Sensory affordance	Design feature that helps users with their sensory actions: seeing, hearing, and feeling (and tasting and smelling) things	A label font size large enough to be discerned
Functional affordance	Design feature that helps users employ a product or system to accomplish work (i.e., usefulness of a system function)	The internal system ability to sort a series of numbers (invoked by users clicking on the Sort button)
Emotional affordance	Design feature that adds emotional impact to the user experience and helps users appreciate and enjoy the interaction	Beautiful aesthetics on a webpage, something that makes interaction fun

Table 27.1 contains a summary of these affordance types and their roles in UX design.

Fig. 27.3 illustrates these different types of affordances in the context of using a doorknob to open a door.

For simplicity of understanding, in this chapter we separate the affordance types into their respective roles. As *Gaver (1991)* says, thinking of affordances in terms of design roles “allows us to consider affordances as properties that can be designed and analyzed in their own terms” (p. 81). Each type of affordance uses different mechanisms, corresponds to different kinds of user actions, and has different requirements for design and different implications in evaluation and problem diagnosis. But, during interaction, they all work together and (in good designs) are integrated, each to mediate the others (*Obrist et al., 2016; Rosenblum, 2013*).

As an example, to get you started, consider a button available for clicking somewhere on a user interface (UI). Sensory affordance helps you sense (in this case, see or notice) it. Sensory affordance can be supported in design by, for example, the color, size, and/or location of the button. Cognitive affordance helps you understand the button by comprehending what the button is used for (in this case, via the meaning of its label). Physical affordance helps you reliably click on this button, so its design support could include the size of the button or its distance from other related buttons.

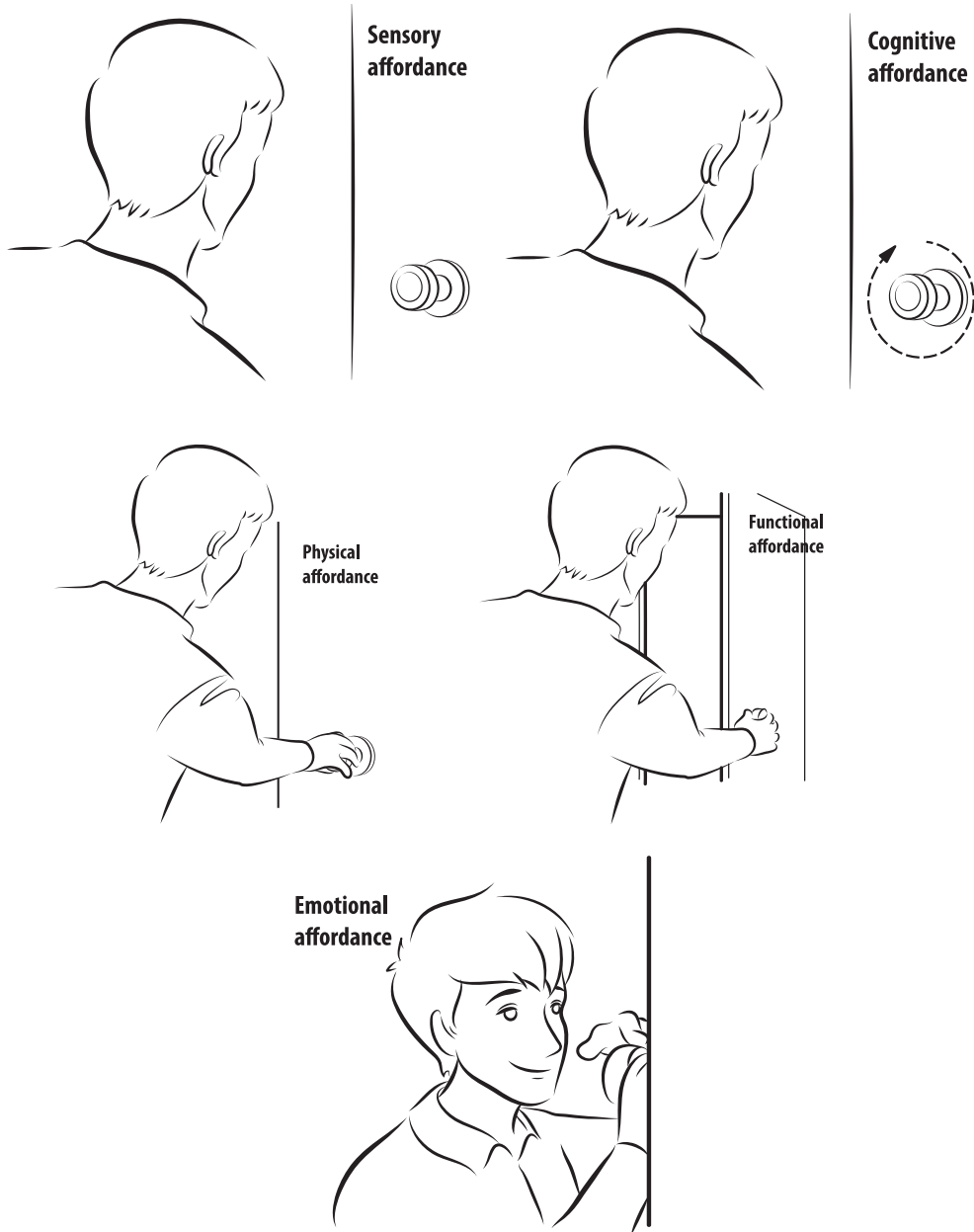


Fig. 27.3

Illustrating the kinds of affordances in UX design.

27.2 COGNITIVE AFFORDANCES

27.2.1 Definition of Cognitive Affordance

A cognitive affordance is a design feature that helps, aids, supports, facilitates, or enables thinking, learning, understanding, and knowing about something. Accordingly, cognitive affordances are among the most significant user-centered design features in present-day interactive systems, screen based or otherwise. They are the key to answering Norman's (1999) question on behalf of the user: "How do you know what to do (and how to do it)?" (p. 39). Knowing what actions to take to respond to the environment and to predict outcomes of actions is foundational to our cognition and operation of the human brain (Buzsáki, 2022).

Cognitive affordance is associated with the semantics or meaning of UI artifacts. As a simple example, the symbol of an icon that clearly conveys its meaning could be a cognitive affordance enabling users to understand the icon in terms of the functionality behind it and the consequences of clicking on it. Another cognitive affordance may be in the form of a clear and concise button label, where the cognitive affordance is used as feed-forward (see [Section 29.4](#)). Cognitive affordances contain knowledge given to the user to predict the outcome before making an action on an object, such as a button, icon, or menu choice, that invokes a certain functionality.

Another use of cognitive affordance is in feedback (see [Section 29.9.1](#))—helping a user know the outcome of a user action taken (e.g., to know what happened after a button click and execution of the corresponding system functionality). Feedback is a cognitive affordance that helps users in knowing whether the course of interaction has been successful so far or whether an error has occurred.

Cognitive affordances play a starring role in UX design, especially for less-experienced users who need help with understanding and learning.

27.2.2 How Do Users Acquire Cognitive Support Information?

The whole point of cognitive affordances is that users need knowledge of how things work. What are the ways users can get that needed information about how to use or operate an object or a device?

Knowledge in the head. First, there is knowledge in the head of the user—it gets there by learning through training or experience.

Knowledge in the world. Then there is knowledge in the world—knowledge inherent in objects and devices that gives us clues as to their operation.

This is the inherent kind of affordance in Gibson's (1977) ecological view: Affordances are the properties and objects of the environment as reckoned relative to a person perceiving the affordance, and "the 'values' and 'meanings' of things in the environment [that] can be directly perceived" by the person. Here, Gibson is talking about physical properties, giving an example of how a horizontal, flat, and rigid surface can be perceived as something that affords support for a person for standing or walking.

Knowledge in the design. Finally, there is knowledge that designers add by way of deliberately designed cognitive affordances, such as a label on a button or a message suggesting an action. This communication depends on a shared understanding between designer and user. See the next section for how we extract meaning from shared cultural conventions.

Example: Mail Icon as Cognitive Affordance

The icon in Fig. 27.4 shows an example of a graphical cognitive affordance that represents knowledge in the world. This icon is a clear and succinct way to use a simple image to convey the meaning behind the corresponding key—this is how you get email. But, as we have said, that message is understood only if both designer and user share the same meaning for the icon.



Fig. 27.4
Clear email icon.

27.2.3 The Meaning of Cognitive Affordances as Found in Shared Conventions

An important function of cognitive affordance is communication from designer to user, which depends on agreement about meaning, but often the meaning of a symbol as a cognitive affordance is not intrinsic to its appearance. The symbols themselves may have no inherent meaning. For

example, the two parallel vertical bars in the symbol for “Pause” on most media-oriented devices, such as a video player, do not contain anything that intrinsically says “pause.” A shared convention about the meaning of an image, icon, symbol (e.g., the image of the pause icon or the octagonal shape of a stop sign), or even some words or phrases can convey that meaning. The image of an envelope shown on the email key in [Fig. 27.4](#) refers to a means of communication, but maybe less so to younger people who have only known electronic versions of communication.

27.2.3.1 Devices for opening doors

In the tradition of *The Design of Everyday Things* (Norman, 1990), we illustrate with a simple and ubiquitous non-computer device, a device for opening doors. The hardware store carries both round doorknobs and lever-type door handles. The visual design of both kinds conveys a built-in cognitive affordance, helping users think or know about usage through the implied message their appearance gives to users: “This is what you use to open the door.” The doorknob and lever handle each suggests, in its own way, the grasping and rotating required for operation.

In cases where users must just push on the door to open it, designers often help users understand this by installing, for example, a brass plate to show that one should push and where to push. Even though this plate may also help avoid handprints on the door, it is a cognitive affordance and not a real physical affordance because it adds nothing to the door itself to help the user in the physical part of the pushing action. Sometimes the word “push” is engraved in the plate to augment the clarity of its meaning as a cognitive affordance.

Perhaps it is mainly because of shared conventions that the message of how to operate a door is so universally understood. Some people may find nothing intrinsic in the appearance of a doorknob that necessarily conveys this information. On another planet, it could seem mysterious and confusing, but for us a doorknob is an excellent cognitive affordance because almost all users share the same easily recognized convention.

Exercise 27.1 Understanding Meaning Based on Cultural Conventions

Can you think of any situation when the meaning of an object, according to shared conventions, is taken for granted in one culture but either has no meaning or carries a completely different meaning in another culture? How would these differences affect the behavior of people from the other culture?

27.2.4 Cognitive Affordance Content and Meaning Design Issues

Of all affordances, cognitive affordances are probably the most important in UX design and may require the most careful design. Content and meaning of cognitive affordances constitute the knowledge that must be conveyed to users to be effective in helping them think, learn, and know what they need to make correct actions. Guidelines for designing the content and meaning of cognitive affordances are based on the following factors:

- Clarity
- Precise wording
- Completeness of information
- Consistency
- User or usage centeredness
- Cognitive directness
- Distinguishability between meanings
- Complexity control

These cognitive affordance design issues and more will be covered in the UX design guidelines of [Chapter 29](#).

27.2.4.1 Cognitive affordance to get the user started

In [Fig. 27.5](#), we show an example of a simple cognitive affordance that helps early users of Keynote on the Mac know what to do to make a slide for presentation. This application, like its Windows counterpart PowerPoint,

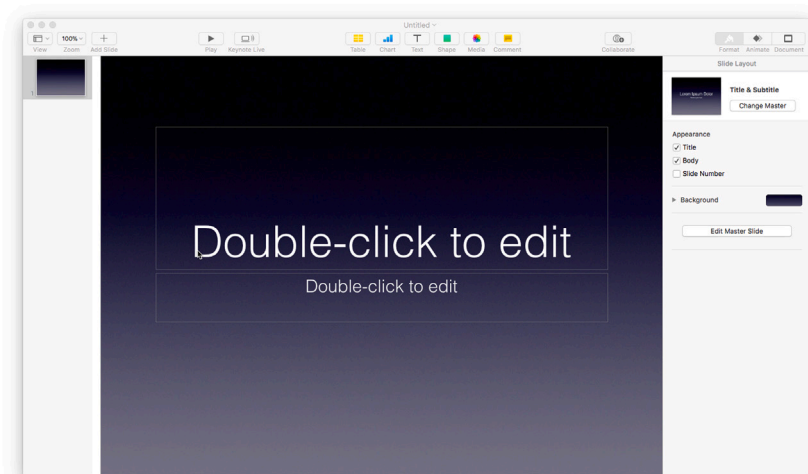


Fig. 27.5
A cognitive
affordance to help
users get started.

is typical of large applications where users often start out by facing a blank screen. All the power of the application is sitting there, but users do not know how to get started. Once users begin, it gets easier because they can respond to feedback in the task thread.

27.2.4.2 Cognitive affordance to help users avoid task completion errors

Sometimes cognitive affordances can help users avoid task completion errors, which are often in the form of a simple reminder to do the last step, such as “Please take your tickets and your receipt.” Fig. 27.6 shows an example of a cognitive affordance as a reminder to include an email attachment.

If any variation of the word “attach” appears in an email, but it is sent without an attachment, the system asks if the sender intended to attach something.

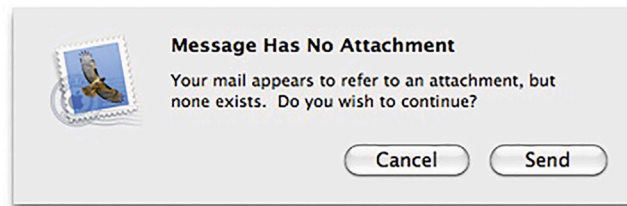


Fig. 27.6

Cognitive affordances used to remind the user to add an email attachment.

27.2.4.3 False cognitive affordances misinform and mislead

There are also things that should not be in a UX design. Among these are false cognitive affordances—things that look like cognitive affordances but are wrong and misinform, mislead, and confuse users. Because of the power of cognitive affordances to influence users, their misuse in design can be a force against usability and a good user experience. Gibson (1977) calls this “misinformation in affordances.” As an example, consider a glass panel that appears to be a door but does not afford passage. Draper and Barton (1993) call this an “affordance bug.”

Example: False Cognitive Affordance in a Door Sign

What do you think about the picture of a sign on the door of a local store shown in Fig. 27.7?

The explanation was that you should use this door only to enter from the outside and not to exit through the door from the inside. They were probably reusing an available design object, the “Do Not Enter” sign, instead of tailoring a sign more specific to the usage situation. The resulting mashup was nonsense.



Fig. 27.7

This door with a confusing sign contains conflicting cognitive affordances.

Example: False Cognitive Affordance in Links Masquerading as Buttons

A common example of a false cognitive affordance is seen in webpage links that are made to look like buttons, but they do not behave as such. Here's one we found in an evaluation of a real digital library system. The gray background of the text in the top menu bar of a digital library website (Fig. 27.8) makes them seem like buttons acting as tabs from which to pick high-level functionality.

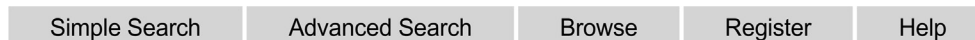


Fig. 27.8

False cognitive affordances in a menu bar with links that look like buttons.

In fact, however, underneath they are only textual hyperlinks, and the gray box around them is only graphical. We still see this often in websites. A user may click on the background, assuming it is part of a button (something we saw more than once in our evaluations), and not get any result. When nothing happens, the user is temporarily confused. Because the button itself is only a text holder, the user must click exactly on the text for functionality. This is a false cognitive affordance in that the button looks like something it is not.

Example: False Cognitive Affordance in a Line That Looks Like the End of a Webpage

Below-the-fold issues on webpages can be compounded by having a horizontal line that happens to fall at the bottom of a screen. Users see the line and assume falsely that it is the bottom of the page and therefore do not scroll, missing possibly vital information below that point.

Example: False Cognitive Affordance in a Microwave Control

As another example, [Fig. 27.9](#) shows a close-up of part of an old microwave.



Fig. 27.9

Useless dial marks between power settings on a microwave.

The dial marks for power settings between the settings for DEFROST and COOK seem to indicate a range of possible settings but, in fact, it is a binary choice: either DEFROST or COOK, with nothing in between. Lots of microwaves have many different settings for the power level between defrosting and full-on cooking, but this one does not. The designer could not resist the temptation to fill in the space between these choices with misleading design details that are false affordances.

27.3 PHYSICAL AFFORDANCES

27.3.1 Definition of Physical Affordance

A physical affordance is a design feature that helps, aids, supports, facilitates, or enables doing physical actions, usually to manipulate UI objects.

Adequate size and an easy-to-access location could be physical affordance features of an interface button design, enabling users to click easily on the button. Physical actions include clicking, grabbing, dragging, touching, and swiping, among others. Physical affordances are associated with operability characteristics of UI artifacts and related human factor issues. These artifacts can include software objects in computer interfaces or they can be real hardware buttons, knobs, handles, dials, or levers. They can be elevator or ATM buttons, or they can even be mechanisms for driving a car, such as a steering wheel or brake pedal.

We usually treat active interface objects on a screen, for example, as physical objects because they can be on the receiving end of physical actions such as clicking or dragging.

Example: Add to Cart Button as a Physical Affordance

Fig. 27.10 provides an example of a physical affordance, the Add to Cart button on [Amazon.com](https://www.amazon.com).

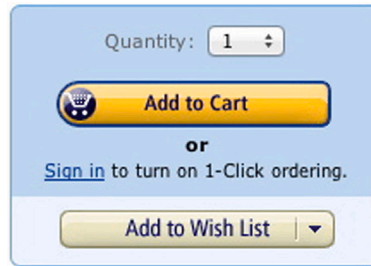


Fig. 27.10

Add to Cart button as physical affordance.

When we analyze the button as a physical affordance, we ask, “How easy is it to click on this button?” The answer, of course, is it is *very* easy to click on this button—after all, they want you to add as much as possible to your cart.

27.3.2 Starring Role in UX Design for Experienced or Power Users

As we said earlier, cognitive affordances play a starring role for novice users. Physical affordances also play a starring role, only for experienced or power users. Their productivity and task performance are all about the speed of physical actions, being able to make physical actions efficient—as rapid and as error-free as possible. Because power users already understand what actions to make, they have less need for cognitive affordances.

27.3.3 Some Physical Affordances Are Better Than Others; Some Depend on the User

The way you decide if one physical affordance is better than another can depend on the individual user. Fig. 27.11 shows the old BlackBerry phone, one



Fig. 27.11

The BlackBerry mobile phone with a physical keypad.

of the few smartphones that had a full hardware keyboard. Some people just liked having a real keyboard like this, one that gives tactile feedback. That is one reason why the makers of BlackBerry kept it in their design for a long time.

On a small device, however, the keys are tiny and close together. For some, this crowded design made for less-than-desirable physical affordance and led to multiple typing errors.

In contrast, other mobile devices, such as the iPhone or iPod, have soft touch keypads (Fig. 27.12). Although some users did not like the design at first, in no time they got used to it. The design earned acceptance through its good visual presentation and audible feedback: “I can see which key I am touching, and I can see and hear when contact is made, which helps me avoid errors.”

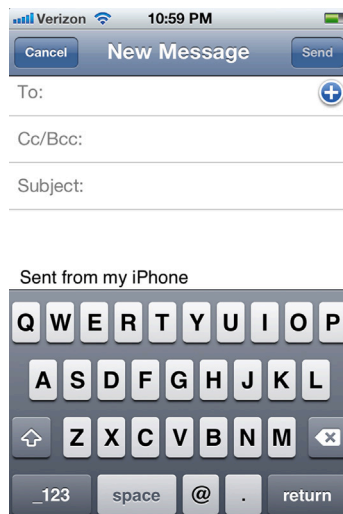


Fig. 27.12
Soft keypad on an old
iPhone.

27.3.4 Physical Affordances for Opening Doors

Here we return to devices for opening doors. As we look at these in Fig. 27.13, we see that each one is different from the others—different in appearance and even different in how they operate. Each one has its own way to afford physical grasping and rotating for door operation.

Some designs, such as a lever, are considered to give better physical affordance than that of a round knob because the lever is easier to use with slippery hands or with an elbow when the hands are full. The push bar on double doors is another example of a physical affordance helpful to door users with full hands.

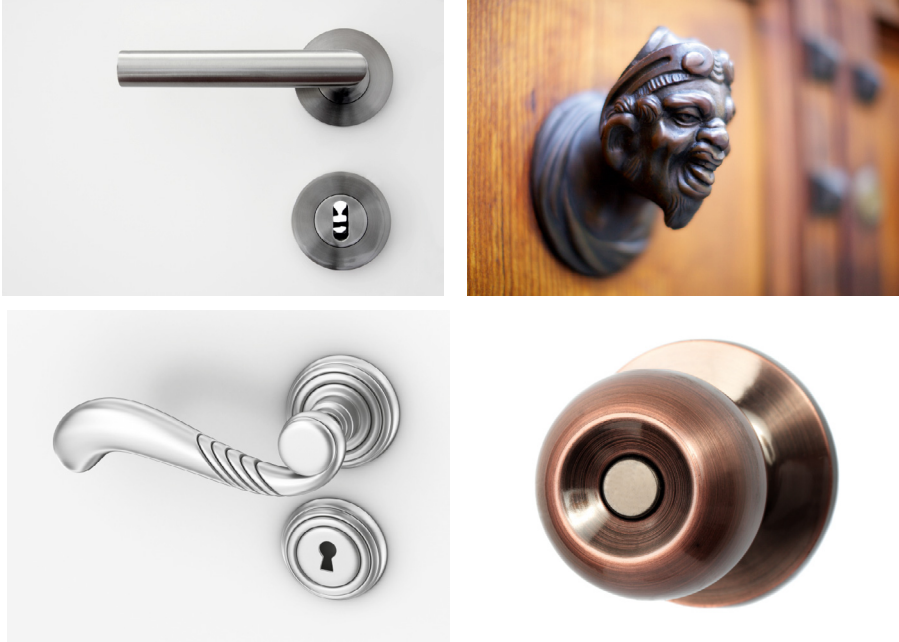


Fig. 27.13
Physical affordances for
opening doors.

Sometimes, the physical affordance to help a user open a door is provided by the door itself; people can open some swinging doors just by pushing on the door. Similarly, sometimes the user of a swinging door must open it by pulling. The door itself does not usually offer sufficient physical affordance for the pulling action, so a pull handle is added.

27.3.5 Physical Devices Can Also Offer Cognitive Affordance

This kind of physical affordance can also act as a cognitive affordance. As you look at it, its appearance may suggest something about how to operate it. It has its own cognitive affordance built into its physical presence. This is an ecological cognitive affordance, in the J. J. Gibson sense, because the lever and the knob in [Fig. 27.14](#) in their natural state (e.g., without a label) both call out, “Grab me, twist me; this is how you open the door.”

27.3.6 Physical Devices Can Also Offer Emotional Affordance

Some designs really look good, sometimes so much so that the appearance is a source of aesthetic pleasure and pride. Lots of people like the handle in [Fig. 27.15](#) because it is a simple design with beautiful sweeping lines.

It’s up to you whether you think the knob in [Fig. 27.16](#) is aesthetic (or grotesque). Either can have emotional impact.

Fig. 27.14

Physical devices as cognitive affordances for how to open a door.



Fig. 27.15

An aesthetic door lever (according to the author).



Fig. 27.16

Does this doorknob elicit feelings?



27.3.7 Physicality

Physicality is a term referring to real direct physical interaction with real physical (hardware) devices, such as in the grasping and moving of knobs and levers. This is not about clicking on soft touch display controls, such as images

of arrows, buttons, or sliders (e.g., to tune a radio or adjust the volume), but about pushing, pulling, grasping, and/or turning real hardware devices, such as knobs, buttons, and levers.

Physicality has been an issue for human factor engineers for a long time. Norman (2007a) brought it to the attention of the HCI/UX community.

Example: Physicality in a Gear-Shift Knob

Fig. 27.17 shows a gear-shift knob that is an archetypical example of physicality.



Fig. 27.17

A car's gear-shift knob that shouts physicality.

The knob itself looks graspable. It is positioned to fit the driver's hand nicely. The leather exterior gives it a solid texture for grasping. Almost everyone likes the kind of physicality that this knob affords. It provides a sure grip and a satisfying feeling of control while shifting confidently through the gears.

Example: Physicality in Radio Controls

Fig. 27.18 shows a car radio that illustrates the issue of physicality.



Fig. 27.18

A car radio with limited physicality.

Although this radio has a physical volume control (the round knob in the center), there is no knob for tuning. Tuning is done by pushing the up and down arrows on the left-hand side. This design lacks the satisfying kind of physicality you get from the almost universally preferred motion of grasping a tuning knob.

27.3.8 Manual Dexterity and Fitts Law

Among the characteristics that affect physical affordances, especially involving object manipulation in graphical user interfaces, are the size and location of the object to be manipulated. A large object is obviously easier to click on than a tiny one, and location of the object can determine how easy it is to reach and manipulate it.

These relationships are represented by Fitts law, an empirically based theory expressed as a set of mathematical formulae that govern certain physical actions in human behavior. As it applies in HCI/UX, Fitts law governs physical movement (e.g., of the cursor) for object selection, moving, dragging, and dropping.

The law is specifically about movement from an initial position to a target object at a terminal position. The formulae predict the time to make a movement as:

- Proportional to $\log_2$ of distance moved
- Inversely proportional to $\log_2$ of target cross section normal to the direction of motion

Also (not expressed by Fitts law), the time to make an error-free landing is inversely proportional (up to a point) to the depth of the target along the path of movement.

The potential for errors is the same—namely, proportional to $\log_2$ of the distance moved and inversely proportional to $\log_2$ of target cross section normal to direction of motion.

Fitts law is one element of HCI theory and is the subject of numerous empirical studies in early HCI literature (Fitts, 1954; MacKenzie, 1992).

27.3.9 Physical overshoot

27.3.9.1 What is physical overshoot?

Physical action overshoot is what occurs when you move an object, cursor, slider, lever, or switch and you go too far in making the physical action,

beyond where you intended. An example in a computer interface is the setting of a slider bar when you pull the slider too far.

Example: Downshifting an Automatic Transmission

Fig. 27.19 shows the transmission gear indicator in an old pickup truck. Here you can see, from right to left at the bottom, the common linear progression of gears from low (number 1) to high (circled D, meaning drive or overdrive).



Fig. 27.19
Gear shifting with an
automatic transmission.

For general driving, the gear progression is a pretty good design, but when coming down a long hill, you may want to downshift to third gear on the fly, using engine drag to maintain the speed limit without wearing out the brakes.

However, because the shifting movement is linear, when you pull that lever down from D, it is too easy to overshoot third gear and end up in second. The result of this error becomes immediately obvious from the sound of the engine straining at high RPMs.

Exercise 27.2 Other Examples of Physical Overshoot

Can you think of any other examples of physical overshoot that you may encounter in your daily life?

27.3.9.2 Constraining physical actions to prevent overshoot

In non-computer examples, you can design to control overshoot by building in physical constraints, as in the next example. In computer applications, you

can ameliorate the risk of overshoot with additional controls, such as using arrow keys to make final slide adjustments.

Example: Constraining a Gear-Shift Pattern

Here is an example of introducing a barrier that blocks physical movement from going too far, in the shifting mechanism of a Toyota Sienna van (Fig. 27.20).



Fig. 27.20

Constrained gear shifting in a Toyota Sienna van, helping prevent physical overshoot.

As an example, say that, while in Drive (4-D in the figure), the driver pulls the lever down to downshift. The lever moves easily down to third gear (3 in the figure) but is then blocked by the cutout template surrounding the shift lever, preventing it from overshooting and going down further to second gear. It is unlikely that a user will accidentally shift into second gear because that requires an extra action of moving the lever to the right and then further down.

This feature is a case of consciously taking physical affordance into account in a design by using constraints on physical actions that help users stay within physical boundaries.

27.3.10 Physical Affordance Design Issues

Physical affordance design issues have to do with the following:

- Help users perform physical actions
- Accommodate physical disabilities
- Avoid awkwardness of physical actions

- Support human factors and ergonomic issues of device design
- Encourage physicality
- Aid in manual dexterity and leverage Fitts law (Fitts, 1954; MacKenzie, 1992)
- Avoid the special characteristic of physical actions called physical overshoot

These physical affordance design issues and more are covered in the UX design guidelines of [Chapter 29](#).

27.4 SENSORY AFFORDANCE

27.4.1 Definition of Sensory Affordance

A sensory affordance is a design feature that helps, aids, supports, facilitates, or enables a user in sensing (e.g., seeing, hearing, touching, feeling) an interaction object, such as a cognitive, physical, or emotional affordance. Design issues for sensory affordances are about the presentation of UI objects and include noticeability, discernibility, legibility (in the case of text), and audibility (in the case of sound) of features or devices associated with visual, auditory, tactile, or other sensations. You cannot use an interaction object if you cannot see it or do not notice it. Presentation of an interaction object is about how the object appears to users, not, for example, how the object conveys meaning or helps do physical actions.

Although cognitive affordance and physical affordance are the stars of UX design, sensory affordance plays an important supporting role. As an example, the legibility of label text on a button is supported by an adequate size font and appropriate color contrast between text and background, so the label can be read.

27.4.2 Design of Sensory Affordances in Support of Other Affordances

Sensory affordances are used as support affordances in the design of presentation of cognitive affordances and physical affordances. For cognitive affordances and physical affordances to serve their respective purposes, they must be sensed (seen, heard, or touched) by users, which means the presentation and appearance of cognitive affordances and physical affordances are about sensory affordances.

27.4.3 Visual Sensory Affordance Design Issues

Visual sensory affordance design issues are about characteristics relating to the user's ability to sense visually (i.e., to see), which include:

- Visibility
- Noticeability
- Discernibility
- Text legibility
- Distinguishability
- Color
- Presentation timing

Next, we present a few examples of these concerns. Other sensory affordance design issues are covered in the UX design guidelines of [Chapter 29](#).

27.4.3.1 Sensory affordance visibility example: Store user cannot find the deodorant

The setting for this non-computer example is a local store, where a shopper has the simple task of buying deodorant. The store had been reorganized, so he could not rely on his memory to locate the deodorant. He went through the aisles, looking at overhead signs but nothing matched his search goal.

Upon asking the clerk, he was told, “Oh, it is right over there,” being pointed to one of the upfront aisles that he had just scanned. “But I don’t see any sign for deodorant.” “Oh, yeah, there is a sign,” the clerk replied. “You just have to get up real close and look right behind that panel on the top of the end shelf.” [Fig. 27.21](#) shows what that panel looked like to someone scanning these upfront aisles.

In [Fig. 27.22](#), you can see the “deodorant” sign, revealed if you “just get up real close and look right behind that panel on the top of the end shelf.”

The panels on top of the shelves are an aesthetic touch, but they were put in a location that exactly blocked the “deodorant” sign and others, rendering that important cognitive affordance invisible from most perspectives in the store.

27.4.3.2 Sensory affordance discernability example: Black-on-black in stereo

[Fig. 27.23](#) shows a picture of a panel on a CD player. The controls, such as for play and stop, are black buttons embossed with black icons. Some designer must have thought it would be cool and aesthetic to use this black-on-black color scheme.



Fig. 27.21

Aesthetic panel blocks visibility of sign as cognitive affordance.



Fig. 27.22

The sign is visible if you look carefully.

These symbols are neither easily discernible nor visually appealing, even in a good light. The symbols are also not raised high enough to be tactile inputs.

This is also a good example about designing for usage in context. Often you want to lower the lights and kick back while listening to music. While this



Fig. 27.23

*Fashionable (?) design
of black icons on a black
background.*

may make for a relaxing ambience, it does nothing to help the visibility and discernibility of the symbols on these controls.

27.4.3.3 Sensory affordance distinguishability example: *Here's soap in your eyes*

In the context of taking a shower, consider the two bottles in Fig. 27.24, one for shampoo and one for conditioner. These are for different purposes but they are so strikingly similar that users cannot easily tell them apart.



Fig. 27.24

*Bottles for shampoo and
conditioner are difficult to
distinguish.*

The important distinguishing labels, for shampoo and for conditioner, are hidden within other text and are in a very small font—indistinguishable, especially for users with soap in their eyes. Remember, too, that users who need eyeglasses cannot wear them in the shower.

So, users sometimes add their own affordances (Section 27.8). In the case shown in Fig. 27.25, a user added labels to the tops of these bottles to tell shampoo from conditioner in the shower.

Fig. 27.26 shows an example of a kind of shampoo bottle design that would have avoided the problem in the first place.



Fig. 27.25

Because of poor sensory affordance, a user had to mark the bottles to tell them apart.



Fig. 27.26

A better design to distinguish the bottles.

In this clever design, the shampoo, the first one you need, is right-side up, and the labeling on the conditioner bottle, the next one you need, is printed so that you stand the bottle “upside down.”

27.4.3.4 Sensory affordance presentation timing example: Just-in-time towel dispenser message

Fig. 27.27 shows a paper towel dispenser in a public bathroom. The next available towel is visible, and the cognitive affordance in the sketch on the cover of the dispenser clearly says, “Pull the towel down with both hands.”



Fig. 27.27

The primary cognitive affordance for taking a paper towel.

But what if the next towel fails to drop down and users cannot grab it? Now a different user action is needed to get a towel, so a different cognitive affordance is required, as illustrated by the push button in [Fig. 27.28](#).



Fig. 27.28

The backup cognitive affordance to help load a new paper towel.

Designers provided this new cognitive affordance, telling the user to push the lever to get the next towel into position. Normally this second cognitive affordance is obscured by the towel, but it becomes visible just when needed.

From the wear marks on the push bar, you can see that this so-called exception case, and the need for this added cognitive affordance, occurs quite often.

27.4.4 Auditory Sensory Issues

The majority of objects in UX designs are visual, but there are also auditory signals to users. Auditory sensory design issues are somewhat parallel to and similar to the visual sensory design issues, only they have to do with sound and how well users are able to hear sounds. Here is a summary of auditory issues for UX design:

- Audibility issues (counterpart to visibility issues)
 - Can users hear the sound?
- Noticeability of audio cues
 - Volume of auditory cues (a major factor in noticeability)
 - Plus, ability of user to control volume
- Intelligibility, discernibility, distinguishability of audio
 - Has much to do with audio quality
 - Can the user make out what is being said?
- Frequency, pitch, composition (of sound)
 - Verbal messages should be centered in the mid-range of human hearing
 - Garbling
 - Interference from background noise
 - Timing of audio cue
 - Tone of audio cues
 - Gentle tone can encourage, reduce stress
 - Harsh tone can annoy, irritate
 - But harsh tone can be used to get attention
- Multiple audio cues can fragment user attention
- Like visual branding, some audio can be subject to branding constraints

Example: Iconic Audio

Interaction designs involving auditory inputs and outputs are becoming more popular. Patel and Hughes (2012) make the case for moving beyond purely visual UIs to distribute tasks over other senses. They propose using auditory channels, in particular, to support understanding of meaning and for decreasing cognitive load. One major advantage of auditory interaction components is that sound usually does not distract user attention from the visual channel.

Patel and Hughes (2012) showed that non-speech sound, what they call “sonification,” is feasible as an interaction channel in certain situations. Making the sounds meaningful entails designing them to be distinguishable in terms of pitch, intensity, and tempo.

27.4.5 Haptic and Tactile Sensory Issues

Haptics is about the sense of touch and physical grasping. Issues of haptics and touch have been around in UX design discussions for a long time and, like auditory interactions, they are being used more often. An interesting example of one such design is the touchable virtual object, as manifest in a touchable hologram (Kugler, 2015). In this approach to haptic interaction, ultrasound is used to focus a shape that can be felt by the human hand. Another pervasive example is the use of haptic feedback on modern smartphones where subtle vibrations communicate user input is received.

Haptic and tactile issues (about touch and feel) are somewhat parallel and similar to visual and auditory issues, including:

- Noticeability: Is a tactile signal strong enough for users to feel it?
- Discernibility: Can users make out what a given tactile signal is?
- Distinguishability: Are there enough differences between two possibly similar tactile signals for users to tell the difference?
- Consistency: Are tactile and haptic representations consistent in meaning?
- Tone: As with other senses, are tactile and haptic stimuli non-irritating?

The factors of noticeability, discernibility, and distinguishability are often aided through some form of vibration (Thompson & Vandenbroucke, 2015) rather than just tactile pressure. One area where design work using tactile vibration is thriving is automotive UIs (Kern & Pfleging, 2013). Haptic feedback is a good match to the physical user actions needed for driving a car. Various kinds of vibrations felt in the steering wheel, the brake pedal, or the driver's seat allow users to receive feedback without having to look away from the driving task. These are especially effective for alerting the driver of situations needing immediate attention, such as an impending accident or drifting from the lane on a road.

27.5 FUNCTIONAL AFFORDANCE

A functional affordance is a design feature that helps users get work done by connecting physical user actions to system, or backend, functionality. Functional affordances are the reason why users will make an action; they want access to the functionality to do something. A functional affordance connects users to the usefulness component of the user experience via the power of the software behind the UI.

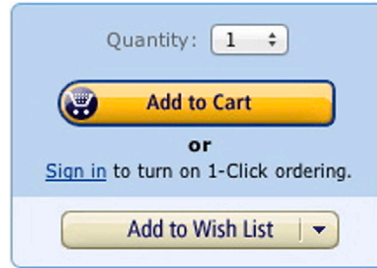


Fig. 27.29

An Add to Cart button viewed as a physical affordance that leads to a functional affordance in the backend software.

For example, if you look at the button in Fig. 27.29 that says Add to Cart, the functional affordance associated with that button is seated in the backend application software that carries out the function implied by that button label—in this case, it puts the item selected for purchase into the virtual shopping cart on [Amazon.com](https://www.amazon.com).

The existence and effectiveness of functional affordances can be a legitimate part of UX evaluation. Beyond that, functional affordances are mainly in the realm of software designers and programmers.

27.6 EMOTIONAL AFFORDANCE

27.6.1 Definition of Emotional Affordance

An emotional affordance is a design feature that helps a user make an emotional connection resulting in emotional impact within the user experience. Emotional affordances include design features that connect to our intuitive appreciation of fun and aesthetics, as well as our sense of identity and challenges to personal growth.

Many UX designs meet all the functional requirements but are not very exciting or interesting. They tick all the boxes but do not light up the lights. The difference is emotional impact, which has become an important component of user experience in most UX designs. The mobile device world is leading attempts to leverage emotional affordances to attract customers. It is especially what new electronic products, such as smartphones and personal music players, are all about. And some have proposed computers and devices that can sense and understand the emotions of users (Thompson, 2010).

Emotional impact is a central concept in the automobile design world. Manufacturers of cars today compete over the pleasure of driving and the fun of using their electronic entertainment systems.

27.6.2 Affordances to Support Meaningfulness

Meaningfulness is a phenomenon that comes out of emotional impact and evolves into a personal relationship that develops and endures over time between a human user and a product that has become part of the user's lifestyle (see [Section 1.4.4](#)). It is difficult to isolate a single design feature that instills meaningfulness; much of that long-term personal feeling of companionship, for example, can be traced to the conceptual design and to the daily usefulness of the product. As an example, many people feel lost without their smartphones in hand.

27.7 PUTTING AFFORDANCES TOGETHER IN DESIGN

User work goals require affordances to work together (Fig. 27.30).

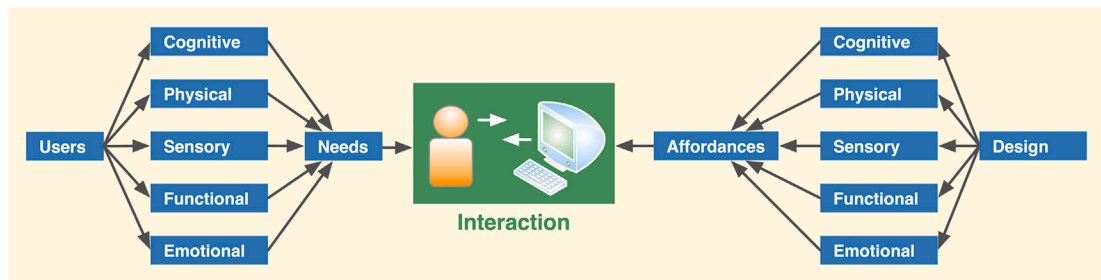


Fig. 27.30

User interaction needs relate directly to affordances in design.

The following are ways that UX design affordances work together to meet user needs:

- Users must understand the interaction objects (through cognitive affordances).
- Users must be able to operate the interaction objects (through physical affordances).
- Users must be able to sense interaction objects (through sensory affordances).
- Users must have access to functionality they need (through functional affordances).
- User would like to engender a good emotional experience in the process (through emotional affordances).

27.7.1 Affordance Roles—an Alliance in Design

In most UX designs, the five types of affordance work together to help users accomplish work goals by sensing, understanding, and using affordances within a UX design. Each kind of affordance plays a different role in

the design of specific attributes of the same artifact, including design of appearance, content, and manipulation characteristics.

An approach to UX design that integrates the different kinds of affordances compels us to consider all five affordance roles together in the design of an interaction artifact.

1. Is the functionality to which this interaction object or artifact gives access useful in achieving user goals through task performance (functional affordance, or purpose of physical affordance)?
2. Does the design include clear, understandable cues about how to use the artifact or about system outcomes if the artifact is a feedback message (cognitive affordance)?
3. Can users easily sense visual (or other) cues about artifact operation (sensory affordance in support of cognitive affordance)?
4. Is the artifact easy to manipulate by all users in the target user classes (physical affordance)?
5. Can users easily sense the artifact for manipulation (sensory affordance in support of physical affordance)?

Considering one affordance role but ignoring another is likely to result in a flawed design. For example, if the wording for a feedback message is carefully crafted to be clear, complete, and helpful (good cognitive affordance), but users do not notice the message because it is displayed out of their focus of attention (poor sensory affordance), or users cannot read it because the font is too small (also poor sensory affordance), the net design is ineffective. A powerful drag-and-drop mechanism on a Mac may offer good physical and functional affordance for opening files (by dragging a file on to an application Icon on the Dock), but lack of a sufficient cognitive affordance to show that it works that way could mean that most users will not use it.

27.7.2 A UX Design Checklist of Affordances

The set of affordances will serve you well as a kind of UX design checklist (Fig. 27.31), against which each part of your design is checked.

- ✓ Functional affordance
- ✓ Cognitive affordance
- ✓ Physical affordance
- ✓ Sensory affordance
- ✓ Emotional affordance

Fig. 27.31
A UX design affordance
checklist.

Example: Affordances for a Sort Capability

This simple example helps you see how the checklist works. Consider the design of a button to invoke, say, some kind of Sort function.

27.7.2.1 Functional affordance considerations

The first item on the checklist is functional affordance. The designer should ask if the intended functionality, the functional affordance, is appropriate, useful, and available to the user. If not, none of the other affordances will be useful in the end. If functional affordance is okay, then we check this item off the list.

27.7.2.2 Cognitive affordance considerations

The next item on the list is cognitive affordance considerations. Is the meaning expressed clearly, in this case, in the label or menu choice that invokes the Sort function in the previous example? Does the label advertise the purpose of the button by ensuring its meaning (in terms of a task-oriented view of its underlying functionality) and is clear, unambiguous, and complete to help the user know when it is appropriate to click on the button?

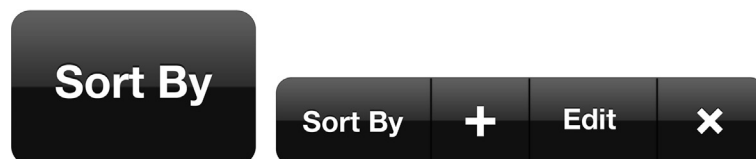
As we say in [Section 29.6.3.10](#), long labels are not necessarily bad. Short labels can be nice, but sometimes it takes more words to convey the full meaning. It is also nice to have both a verb and a noun—to indicate the action and the object. So, instead of just Sort, we could expand it to Sort Records. Sometimes even adding an adjective will help convey the meaning more fully, in this case, Sort Address Records. Or an adverbial phrase may add information about how the sort will be done: Sort By Last Name.

Now, we can check cognitive affordance off the list.

27.7.2.3 Physical affordance considerations

The designer is next led to consider how physical affordance is to be supported in the button design. Does our Sort button design from the previous example provide adequate physical affordance for easy clicking? Is the size of the button large enough to click on it easily? The button at the left of [Fig. 27.32](#) is large enough so that a little accidental cursor movement during

Fig. 27.32
Different buttons have
different physical
affordances.



clicking will not move it off the button. The button on the right, however, is perhaps too small.

Other designer questions to ask include: Is the button located near other artifacts used in the same and related tasks to minimize mouse movement between task actions? Is the button located far enough away from other, non-related UI objects to avoid clicking on them erroneously?

This is also the time to check on physical disabilities. You may not have thought of this, but even some people who do not appear to have physical disabilities may have trouble clicking.

Now, we can check physical affordances off our list.

27.7.2.4 Sensory affordance considerations

Now, the designer is asked to consider sensory affordance in support of both cognitive affordance and physical affordance in the button design in the previous example.

In support of the physical affordance, the button must be a color, size, and shape that make it noticeable and, if possible, it should be located in the screen layout so that it is near enough to the user's focus of attention.

In support of cognitive affordance, the button label must have effective font size and color contrast, for example, to help the user discern the label text to read it.

On the left side of Fig. 27.33, we have a nice Sort By button, but the low-contrast label in the middle is more difficult to read. The font on the right-hand side is unnecessarily small. There's plenty of space; use it!

Now we can check sensory affordances off our list.

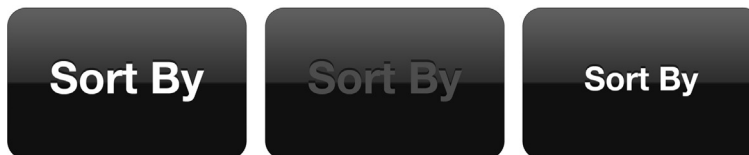


Fig. 27.33
Different Sort By buttons
with different sensory
affordances.

27.7.2.5 Emotional affordance considerations

Finally, we ask about emotional affordances in this design from our previous example. Despite it being difficult to get emotional about a Sort button or function, you could make it a little more fun with an animation that shows how the sort works. Novice users may be intrigued by this. Or maybe you could find some interesting way to use sound. Challenge yourself and be creative!

In any case, let's check this off the list.

Exercise 27.3 Affordance Design Checklist

Now that we have shown you how to use a checklist of affordance types in the consideration of a design for a Sort button, your job in this exercise is to explain how to apply the same checklist when designing feedback when used as an error message in a system of your choice.

27.8 USER-CREATED AFFORDANCES AS A WAKE-UP CALL TO DESIGNERS

A user-created affordance is an affordance that is added to the original design (e.g., by taping on a label) by a user who needs that affordance for a better user experience.

If a device in the everyday world does not suit the user, we will frequently see users modify the apparatus, briefly and unknowingly switching to the role of designer. We have all seen the little cognitive or physical affordances added to devices by users: Post-it notes added to a computer monitor or keyboard or a better grip taped to the handle of something. These trails of user-created artifacts blazed in the process of day-to-day usage are like wake-up messages, telling designers what the users think they missed in the design.

A common example of trails (literally) of user-made artifacts is seen in the paths worn by people as they walk. Sidewalk designers usually like to make the sidewalk patterns regular, symmetric, and rectilinear. However, the most efficient paths for people getting from one place to the other are often less tidy but more direct. Wear patterns in the grass show where people need or want to walk and, thus, where the sidewalks probably should have been located. The rare and creative sidewalk designer will wait until seeing the worn paths, employing the user-made artifacts as clues to drive the design.

Example: Confusing Glass Door

Fig. 27.34 shows a glass door in a convenience store as an example of an added cognitive affordance. The glass and stainless-steel design is elegant; the perfectly symmetric layout and virtually unnoticeable hinges contribute to the uncluttered aesthetic appearance (good emotional affordance), but these same attributes work against cognitive affordance for its operation.

The storeowner noticed many people unsure about which side of the stainless-steel bar to push or pull to open the door, often trying the wrong side first. To help his customers with what should have been an easy task in the first place, he glued a bright yellow cardboard arrow to the glass, pointing out the correct place to operate the door.



Fig. 27.34

Glass door with a user-added cognitive affordance (arrow) indicating proper operation.

Example: Copier Darkness Icon

As another example of user-created affordances, consider the icons shown in Fig. 27.35. These are for lightness and darkness settings on a home office copier; icons with ambiguous meanings. The icon for lighter copies is whiter, but the white can be interpreted as part of the copy, and it seems denser than the icon for darker copies, so the user had to add his own label, as you can see in the figure.

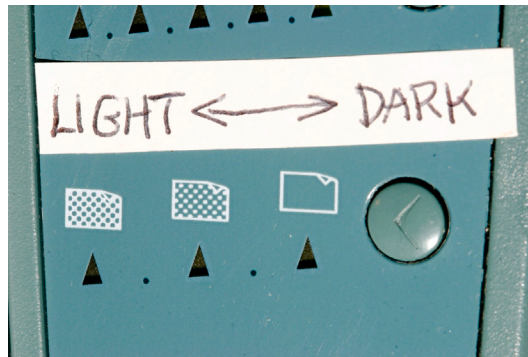


Fig. 27.35

A user-created cognitive affordance explaining copier darkness settings.

These trails of often inelegant but usually effective artifacts added by frustrated users leave a record of affordance improvements that designers should consider for all their users. Perhaps if designers of the everyday things that Norman (1990) discussed had included usability testing in the field, they would have found these problems before the products went to market.

In the software world, most applications have only very limited capabilities for users to set their preferences. Would it be nicer for software users if they

could modify UX designs as easily as applying a little duct tape, a Post-it note, or extra paint here and there?

Example: Car Drink Holder

Fig. 27.36 shows how a car owner created an artifact to replace an inadequate physical affordance—a built-in drink holder that was too small for today’s super-sized drinks. During one trip, the user improvised with a shoe, resulting in this interesting example of a user-installed artifact.



Fig. 27.36

A user-made automobile cup-holder artifact (from Howarth [2002]. Courtesy of Roundel, BMW Car Club Magazine).

Example: Letterhead Printing

Consider a desktop printer used occasionally to print on letterhead stationery. Inserting the stationery on top of the existing plain paper supply in the printer does this rather easily. The only problem is that it isn’t easy to determine the correct orientation of the sheet to be inserted because of the following:

- There is no clear mental model of how the sheet travels through in the interior mechanism of the printer.
- Printers can vary in this configuration.
- The design of the printer itself gives no cognitive affordance for loading a single sheet of letterhead.

Thus the user attached his own white adhesive label (Fig. 27.37). As Norman (1990) says, “When simple things need pictures, labels, or instructions, the design has failed” (p. 9).

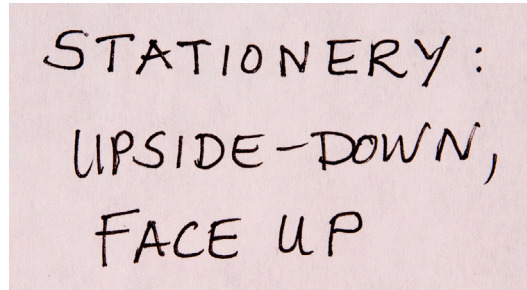


Fig. 27.37

A user-created cognitive affordance to help users know how to insert blank letterhead stationery.

27.9 A LITTLE HISTORY OF THE CONCEPT OF AFFORDANCES

Donald Norman. Norman was the one who introduced the concept of affordances to HCI and UX, using the term in his 1988 book, *The Psychology of Everyday Things*, a book he renamed in 1990 to *The Design of Everyday Things*. He is a psychologist and rightly saw that usage and usability were about psychology, but the real point of his book was design. In any case, Norman was not the first to use the term affordance.

J.J. Gibson. It is always risky to make a claim about who used a particular term first, but we think in this case it was Gibson, more than a decade earlier than when it emerged in HCI. In 1977, he wrote an article on the theory of affordances, and he followed up with a complete book on his ecological approach to visual perception (Gibson, 1979).

As an ecological psychologist, Gibson saw animal perception, especially human perception, tied closely to the setting—to the objects that constitute the environment of the animal and how they offer different capabilities to the animal. So, in this light, an affordance is something in people's environment that helps them do what they need to while they are living in that environment.

One of the simplest examples Gibson gave was to think about the hard surface of the ground. If it can support my weight, then it affords my standing or sitting on that piece of ground.

William Gaver. Beyond Gibson and Norman, Gaver (1991) had an influence on our thinking about affordances as he saw affordances in design as a way of focusing on strengths and weaknesses of technologies with respect to the possibilities they offer to people who use them.

He extended the concepts by showing how complex actions can be described in terms of groups of affordances, sequential in time and/or nested in space. Affordances are revealed over time, with successive user actions, for

example, in the multiple actions of a hierarchical drop-down menu. Gaver (1991) defined his own terms somewhat differently from those of Norman or Gibson.

Joanna McGrenere and Wayne Ho. McGrenere and Ho (2000) also influenced our work, but they used yet a different terminology that they then compared to Gaver’s (1991) vocabulary, demonstrating the need for a richer, more consistent and standard vocabulary. Now, the literature about affordances continues to grow. For example, it was featured prominently in CHI 2012, the annual conference in HCI and UX.

In most of the related literature, design of cognitive affordances (whatever they are called in a given paper) is acknowledged to be about design for the cognitive part of usability—that is, ease of use in the form of learnability for new and intermittent users (who need the most help in knowing how to do something). All authors who write about affordances give their own definitions of the concept, but there is scant mention beyond cognitive affordances to, for example, physical affordance design.

Sensory affordance is neglected even more in the literature. Most other authors include sensory affordance only implicitly and/or lumped in with cognitive affordance rather than as a separate explicit concept. Thus, when these authors talk about perceiving affordances, including Gaver’s (1991) and McGrenere and Ho’s (2000) phrase “perceptibility of an affordance,” they are referring (in our terms) to a combination of sensing (e.g., seeing) and understanding physical affordances through sensory affordances and cognitive affordances.

Gaver (1991) refers to this same mix of affordances when he says, “People perceive the environment directly in terms of its potential for action.” In contrast, our use of the term *sense* has a markedly narrower orientation toward discerning via sensory inputs, such as seeing and hearing.

27.10 SEGUE

In this chapter, we covered the theory of affordances. In the next chapter, we discuss the interaction cycle, a framework to reason about and design for different phases users go through in their use of a product or system.